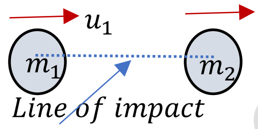
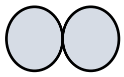
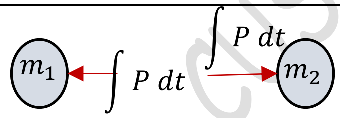
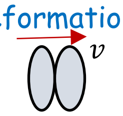
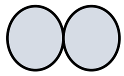
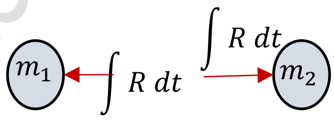
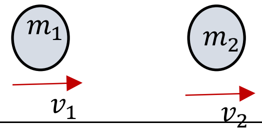
IMPACT

The phenomenon of collision of two moving bodies where active and reactive forces of very large magnitude act for a very short interval of time is called impact. In general, there are two types impact. When the direction of velocities of the bodies are along the line joining their mass centres, it is called a direct impact/ central impact. When the motion of one or both of the particles is at an angle with the line of mass centres, it is called an oblique impact.

Direct/Central Impact

1. Before impact	$u_1 > u_2$	
2. During impact	Deformation phase (A) 	 Deformation impulse
	Maximum deformation (B) 	
	Restitution Phase (C) 	 Restitution impulse
3. After impact		

1. Consider two bodies of masses m_1 and m_2 moving with velocities u_1 and u_2. If $u_1 > u_2$, they will collide with each other.
2. During impact, the bodies will deform. The bodies will undergo a period of deformation such that they exert an equal but opposite deformation impulse $\int P\,dt$ on each other.
   The velocity of m_1 will decrease and that of m_2 will increase, and at the instant of maximum deformation both will have a common velocity v.
   After that, a period of restitution occurs in which the bodies will either regain its original shape or remain permanently deformed. The equal and opposite restitution impulse $\int R\,dt$ pushes the particles apart from each other.
3. After separation, the bodies will move with velocities v_1 and v_2

When we consider the system of two bodies, the deformation and restitution impulses are internal to the system, they are equal and opposite. Hence, by the conservation of momentum

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

To find v_1 and v_2, we need one more equation. That we can find out by considering the impulse-momentum principle of each body.
Consider m_1 in the deformation phase.
Applying the impulse-momentum principle,

$$m_1u_1 - \int P\,dt = m_1v \qquad \ggg\ggg\ggg \qquad \int P\,dt = m_1u_1 - m_1v$$

In the restitution phase

$$m_1v - \int R\,dt = m_1v_1 \qquad \ggg\gg \qquad \int R\,dt = m_1v - m_1v_1$$

$$\frac{\int R\,dt}{\int P\,dt} = \frac{m_1 v - m_1 v_1}{m_1 u_1 - m_1 v} = \frac{v - v_1}{u_1 - v}$$

This ratio of restitution impulse to the deformation impulse is known as the coefficient of restitution e

Similarly let us consider the body m_2

$$m_2 u_2 + \int P\,dt = m_2 v$$

$$m_2 v + \int R\,dt = m_2 v_2$$

$$e = \frac{\int R\,dt}{\int P\,dt} = \frac{v_2 - v}{v - u_2}$$

Eliminating v between the two expressions of e

$$e u_1 - e v = v - v_1$$

$$e u_1 + v_1 = v(1 + e)$$

Similarly,

$$e u_2 + v_2 = v(1 + e)$$

Therefore,

$$\boldsymbol{v_1 - v_2 = -e(u_1 - u_2)}$$

$$Relative\ velocity\ after\ impact = -e(relative\ velocity\ before\ impact)$$

Elastic Impact: When the deformation impulse is equal to the restitution impulse, the collision is said to be elastic. There will not be any deformation of the body and no loss of energy. In such a case, the coefficient of restitution $e = 1$.

Plastic impact: If the restitution impulse is zero, both bodies stick together and move as a single body. Such an impact is known as plastic impact. Here $e = 0$.

Semi-elastic impact: If the value of e is between 0 and 1, the impact is known as semi-elastic impact.

> Example 1
>
> Disk A has a mass of $250\ g$ and is moving on a smooth horiozontal surface with an initial velocity $u_1 = 2m/s$. It makes a direct collision with disk B which has a mass of $175\ g$ and originally at rest. If both disks are of same size and the collision is perfectly elastic, determine the velocity of the disks after collision.

$$m_1 = 250\ g, m_2 = 175\ g, u_1 = 2\frac{m}{s}, u_2 = 0, e = 1$$

Initial momentum of the system $= m_1 u_1 + m_2 u_2 = 250 \times 10^{-3} \times 2 + 0 = 500 \times 10^{-3}\ kg.m/s$

Final momentum of the system $= m_1 v_1 + m_2 v_2 = 250 \times 10^{-3} v_1 + 175 \times 10^{-3} v_2$

By conservation of momentum,

$$500 \times 10^{-3} = 250 \times 10^{-3} v_1 + 175 \times 10^{-3} v_2$$

Also, we have,

$$v_1 - v_2 = -e(u_1 - u_2)$$

$$v_1 - v_2 = -2$$

Solving these two equations, we get

$$v_1 = 0.353\frac{m}{s}$$

$$v_2 = 2.353\frac{m}{s}$$

Example 2

Block A has a mass of $3\ kg$ and is sliding on a rough horizontal surface with a velocity $u_1 = 2\ m/s$ when it makes a direct collision with B, which has a mass of $2\ kg$ and is originally at rest. If the collision is pefectly elastic, determine the velocity of each block just after collsion and the distance between the blocks when they stop sliding. The coefficeint of kinetic friction between the blocks and the plane is 0.3

$$m_1 = 3\ kg, m_2 = 2\ kg, u_1 = 2\ \frac{m}{s}, u_2 = 0, e = 1$$

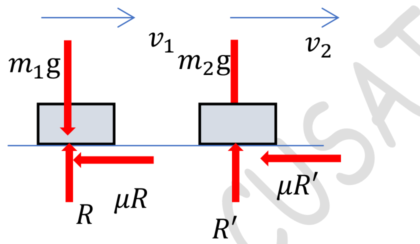
$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$v_1 - v_2 = -e(u_1 - u_2)$$

On solving, we get

$$v_1 = 0.4\ m/s$$

$$v_2 = 2.4\ m/s$$



Let us consider the motion of the bodies after collision

After collision both bodies start moving with velocities $v_1 = 0.4\ m/s$ and $v_2 = 2.4\ m/s$ after travelling a distance x_1 and x_2, they stop.

Consider body 1

Initial kinetic energy of the body $= m_1 \frac{v_1{}^2}{2} = 3\ kg \times \frac{0.4^2}{2} = 0.24\ Nm$

Final kinetic energy of the body=0

Net work done by the forces in the direction of motion

$= -\mu R \times x_1 = -\mu\, m_1 g \times x_1 = -0.3 \times 3 \times 9.81 \times x_1 = \ -8.829\, x_1$ (−ve because force and displacement are opposite)

By work energy principle, change in kinetic energy= net work done

$$0 -\ 0.24\ Nm = -8.829\, x_1$$

$$x_1 = 0.027\ m$$

Consider body 2

Initial kinetic energy of the body $= m_2 \frac{v_2{}^2}{2} = 2\ kg \times \frac{2.4^2}{2} = 5.76\ Nm$

Final kinetic energy of the body=0

Net work done by the forces in the direction of motion

$$= -\mu R' \times x_2 = -\mu\, m_2 g \times x_2 = -0.3 \times 2 \times 9.81 \times x_2 = -5.886\, x_2$$

By work energy principle, change in kinetic energy= net work done

$$0 -\ 5.76\ Nm = -5.886\, x_2$$

$$x_2 = 0.978\ m$$

Therefore, the distance between the blocks when they stop $= x_2 - x_1 = 0.952\ m$

Example 3

A wooden block weighing $45\ N$ rests up on a rough horizontal plane, the coefficient of friction between the two being 0.4. If a bullet weighing $0.25\ N$ is fired horizontally in to the block with a muzzle velocity $v = 600\ m/s$, how far will the block be displaced from its initial position? Assume that the bullet remains inside the block.

$$W_1 = 45\ N, W_2 = 0.25\ N,, u_1 = 0, u_2 = 600\frac{m}{s}, \qquad e = 0\ (plastic\ impact)$$

Since $v_1 - v_2 = -e(u_1 - u_2) \ggg\ggg v_1 = v_2 = v$

Consider the impact of the bullet with the block. From the conservation of momentum,

$$\frac{W_1}{g}u_1 + \frac{W_2}{g}u_2 = \left(\frac{W_1}{g} + \frac{W_2}{g}\right)v$$

$$45 \times 0 + 0.25 \times 600 = 45.25 \times v$$

$$v = 3.32\ m/s$$

Now, consider the motion of the block and the bullet after impact.

The block and the bullet start moving with velocity $v = 3.32\ m/s$ and after travelling a distance x they come to rest.

The initial kinetic energy of the block and bullet

$$= \left(\frac{W_1}{g} + \frac{W_2}{g}\right)\frac{v^2}{2} = \frac{45.25}{9.81} \times \frac{3.32^2}{2} = 25.42\ Nm$$

Final kinetic energy $= 0$

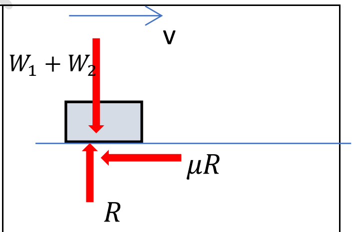
The work done by the forces in the direction of motion $= -\mu R \times x$

But $R = W_1 + W_2$

The change in kinetic energy is equal to the net work done.

$$0 - 25.42 = -\mu(W_1 + W_2) \times x$$

$$x = 1.4\ m$$



Example 4

A pile weighing $4450N$ is driven by dropping a hammer of weight $8900N$ on the top of it from a height of $3m$. If the resistance to penetration is constant and is equal to $266900\ N$, how many blows of the hammer are required to drive the pile to a depth of $0.3\ m$ if the coefficient of restitution is 0.25. Assume that the hammer drops from the same height each time.

Let us first find out the velocity of the hammer just before its impact with the pile. It is being dropped from a height of 3m. Therefore, the velocity before impact,

$$u_1 = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 3} = 7.67\ m/s$$

Velocity of the pile just before impact, $u_2 = 0$

Let $v_1 and\ v_2$ be the velocities of the hammer and pile just after the impact.

$$\frac{W_1}{g}u_1 + \frac{W_2}{g}u_2 = \frac{W_1}{g}v_1 + \frac{W_2}{g}v_2$$

$$8900 \times 7.67 + 0 = 8900v_1 + 4450 \times v_2$$

$$2v_1 + v_2 = 15.34$$

We also have the equation

$$v_1 - v_2 = -e(u_1 - u_2)$$

$$v_1 - v_2 = -0.25(7.67 - 0) = -1.9175$$

On solving, we get

$$v_2 = 6.39\frac{m}{s}, \qquad v_1 = 4.47\ m/s$$

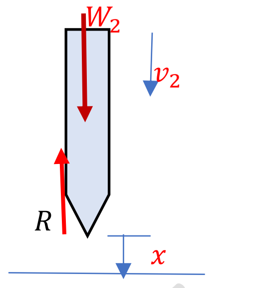
Considering the motion of the pile after the impact. It starts its motion with a velocity $v_2 = 6.39\ m/s$ and comes to rest after penetrating a distance of x.

Initial kinetic energy of the pile $= \frac{W_2}{g}\frac{{v_2}^2}{2} = \frac{4450}{9.81} \times \frac{6.39^2}{2} = 9261.1\ Nm$

Final kinetic energy of the pile$= 0$

Change in kinetic energy of the pile$= 0 - 9261.1 = -9261.1\ Nm$

Net work done by the forces W_2 and $R = W_2 x - Rx = (8900 - 266900)x$



By work-energy principle, change in kinetic energy=net work done

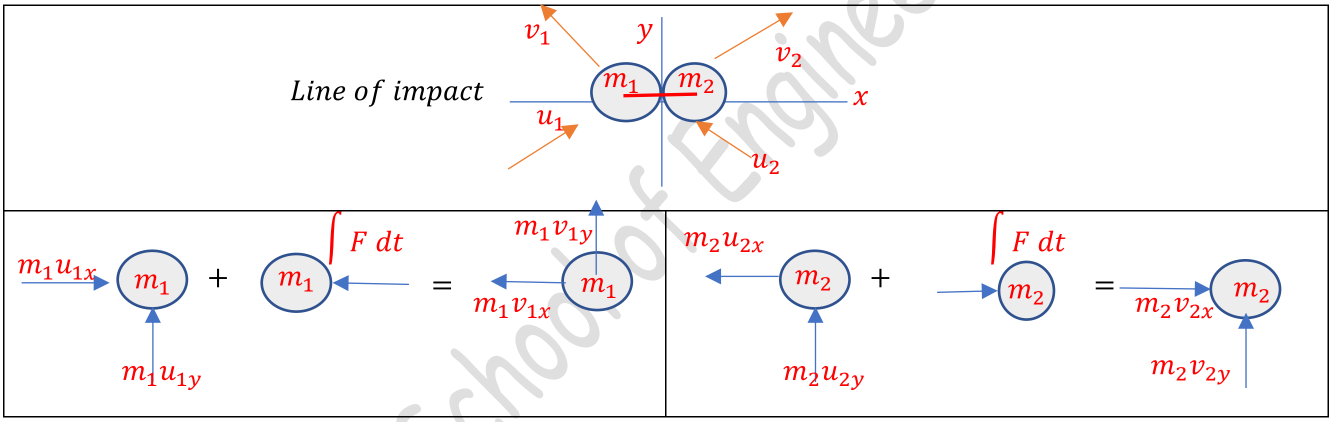
$$-258000x = -9261.1$$

$$x = 0.036\ m$$

That is, the pile penetrates a distance of $0.036\ m$ in a blow. Therefore, the number of blows required for the penetration of 0.3 m$= \frac{0.3}{0.036} = 8.33 \cong 9\ blows$

Oblique impact

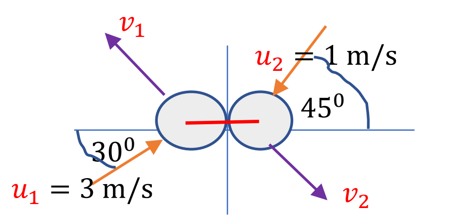
In oblique impact, one or both the velocities will not be along the line of mass centres (line of impact). In such a case, we consider the components of velocities in two mutually perpendicular direction.

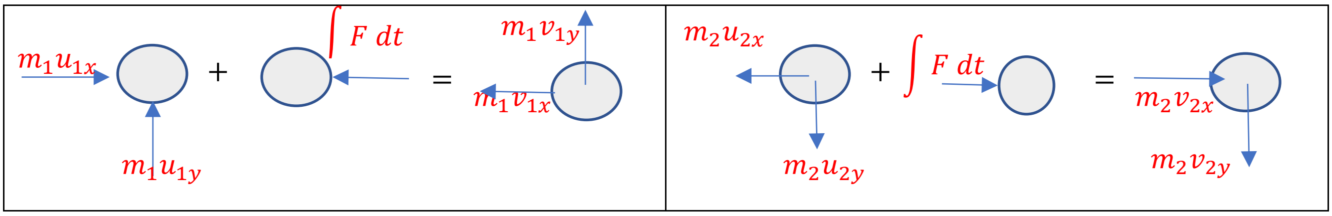


The momenta of the bodies are resolved along the x-y axes. The momentum of the system is conserved along the line of impact	$m_1u_{1x} - m_2u_{2x} = -m_1v_{1x} + m_2v_{2x}$
The coefficient of restitution relates the relative velocity components along the line of impact	$-v_{1x} - v_{2x} = -e(u_{1x} + u_{2x})$
Momenta of the body A and body B are conserved along the y-axis as no impulse act on the A and B along y-direction	

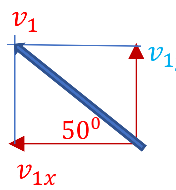
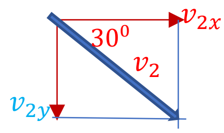
Example 1

Two smooth disks A and B having a mass of $1\ kg$ and $2\ kg$ respectively collide with each other with velocities as shown. If the coefficient of restitution for the disks is $e = 0.75$, determine the velocities of each disk after collision. Neglect friction.





The momenta of the bodies are resolved along the x-y axes. The momentum of the system is conserved along the line of impact	$m_1u_{1x} - m_2u_{2x} = -m_1v_{1x} + m_2v_{2x}$ $1 \times 3\cos 30 - 2 \times 1\cos 45 = -1 \times v_{1x} + 2 \times v_{2x}$ $-v_{1x} + 2v_{2x} = 1.184$
The coefficient of restitution relates the relative velocity components along the line of impact	$-v_{1x} - v_{2x} = -e(u_{1x} + u_{2x})$ $-v_{1x} - v_{2x} = -0.75(3\cos 30 + 1\cos 45) = -2.48$ $-v_{1x} - v_{2x} = -2.48$ Solving, we get $v_{2x} = 1.22\frac{m}{s}, v_{1x} = 1.26\, m/s$
Momenta of the body A and body B are conserved along the y-axis since no impulse act on the A and B along y	$m_1u_{1y} = m_1v_{1y}$ $v_{1y} = u_{1y}$ ⋙⋙ $v_{1y} = 3\sin 30$ ⋙⋙ $v_{1y} = 1.5 m/s$ $m_2u_{2y} = m_2v_{2y}$ $u_{2y} = v_{2y}$ ⋙ $-1 \times \cos 45 = -v_{2y}$ ⋙ $v_{2y} = 0.707 m/s$

$v_{1x} = 1.26\frac{m}{s}$ ← $v_{1y} = 1.5\frac{m}{s}$ ↑	$v_1 = \sqrt{{v_{1x}}^2 + {v_{1y}}^2} = 1.96\frac{m}{s}$ $\tan\theta = \frac{v_{1y}}{v_{1x}}$ ⟪⟫ $\theta = \tan^{-1}\left(\frac{1.5}{1.26}\right) = 50^0$	
$v_{2x} = 1.22\frac{m}{s}$ → $v_{2y} = 0.707\frac{m}{s}$ ↓	$v_2 = \sqrt{{v_{2x}}^2 + {v_{2y}}^2} = 1.41\frac{m}{s}$ $\tan\emptyset = \frac{v_{2y}}{v_{2x}}$ ⟪⟫ $\emptyset = \tan^{-1}\left(\frac{0.707}{1.22}\right) = 30^0$	

#Exercise 1
A ball dropped from rest on to a cement floor rebounds 0.8 times the height through which it fell. Neglecting air resistance, determine the coefficient of restitution.

#Exercise 2
A ball moving with a velocity v collides elastically with another ball at rest. If the weight of ball second ball is twice that of the first, find the velocities of the balls after impact.

#Exercise 3
A car of weight W starts from rest at A and rolls without friction along a 15^0 inclined plane to B where it strikes a block also of weight W initially at rest. Assuming a plastic impact at B, calculate the distance travelled by them along the plane before coming to rest. Take the coefficient of friction between the block and the plane is 0.5.